

07.05 Solving Rational Equations

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Rational Expressions And Equations	State denominator restrictions before simplifying or solving.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Solving Rational Equations

1.

$$\frac{x^2 - 9}{x^2 + 3x}$$

2.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

3.

$$\frac{x + 1}{x^2 - 4}$$

4.

$$\frac{x^2 - 9}{x^2 + 3x}$$

5.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

6.

$$\frac{x+1}{x^2-4}$$

7.

$$\frac{x^2-9}{x^2+3x}$$

8.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

9.

$$\frac{x+1}{x^2-4}$$

10.

$$\frac{x^2-9}{x^2+3x}$$

11.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

12.

$$\frac{x+1}{x^2-4}$$

13.

$$\frac{x^2-9}{x^2+3x}$$

14.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

15.

$$\frac{x+1}{x^2-4}$$

16.

$$\frac{x^2-9}{x^2+3x}$$

17.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

18.

$$\frac{x+1}{x^2-4}$$

19.

$$\frac{x^2-9}{x^2+3x}$$

20.

$$\frac{1}{x} + \frac{1}{2} = \frac{3}{4}$$

B. Apply the key formulas

1.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

2.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

3.

LCD clears denominators

4.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

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$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

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